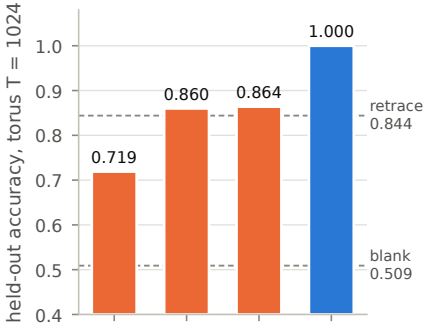
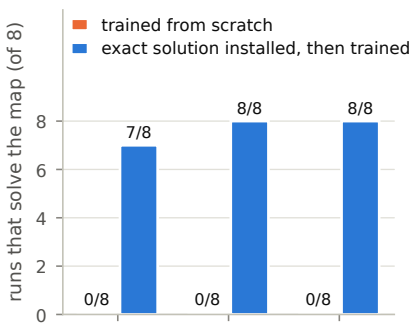
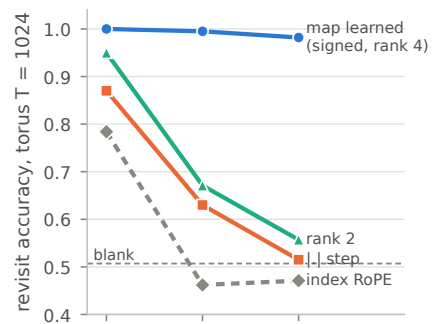

(a) One path layer learns the map; index depth does not
held-out accuracy, torus T = 1024
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.719
0.860
0.864
1.000
retrace 0.844
blank 0.509
index 1 layer
index 4 layers
index loop x4
path 1 layer


(b) The solution exists and training keeps it, but training from scratch does not find it
trained from scratch
exact solution installed, then trained
runs that solve the map (of 8)
8
6
4
2
0
7/8
8/8
8/8
0/8
0/8
0/8
rank 2 (torus)
|.| step (ring)
|.| step (torus)


(c) Every failure keeps short returns and loses long and wrap-around ones
revisit accuracy, torus T = 1024
1.0
0.9
0.8
0.7
0.6
0.5
0.4
map learned (signed, rank 4)
rank 2
|.| step
index RoPE
blank
gap < 128
gap >= 128
wrap-only